

Thermo-Mechanical Description of a Wankel Rotary Engine: Computational simulation of
the thermodynamic cycle and combustion

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Abstract

Este es un breve resumen del documento. Debería incluir una visión general del contenido del artículo en no más de 250 palabras.

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1. Introduction

The energy transformation sector faces significant challenges due to its reliance on conventional technologies, which often exhibit low thermal efficiencies, according to Wang, Shuai, Li, and Yu (2021), and high levels of pollutant emissions. Traditional engines, such as Otto and Diesel cycle engines, frequently fail to optimize the thermal energy generated during the combustion process, leading to substantial energy losses and a considerable environmental footprint.

The use of unconventional technologies, such as Wankel rotary engines, presents an opportunity to rethink energy transformation processes. However, these engines remain underutilized, primarily due to historical concerns regarding their efficiency and durability. Furthermore, detailed analyses that consider both the thermodynamic and mechanical aspects of this technology under real-world conditions are scarce.

In the current context, where maximizing energy efficiency and minimizing environmental impact are priorities, it is essential to assess the potential of the Wankel engine as a viable alternative. Such an analysis would not only deepen the understanding of its behavior under various operational conditions but also provide a robust foundation for its optimization in practical applications, contributing to a more efficient and sustainable energy transformation paradigm.

This work proposes a comprehensive methodology that integrates geometric modeling, thermodynamic cycle simulation, chemical reaction analysis, and vectorial mechanics. One of the main contributions lies in the incorporation of torque analysis based on vector cross-products between the applied force and the effective arm generated by rotor motion. By combining analytical functions—such as the epitrochoidal volume variation—with a dynamic mechanical model, this article offers a deeper and more accurate understanding of the power transmission mechanisms in Wankel engines. This integration enables the

prediction of instantaneous torque, evaluation of performance under transient conditions, and ultimately supports the optimization of design variables with greater precision.

1.1. State of the Art

The Wankel engine is characterized by its innovative design, utilizing a rotor with curved sides that rotates eccentrically within an epitrochoidal-shaped housing. This motion generates three working chambers that alternately perform the four stages of the combustion cycle: intake, compression, combustion, and exhaust. One of its main advantages is its reduced number of moving parts compared to traditional piston engines, which significantly decreases friction and wear (Froede, 1961). Moreover, it operates more smoothly with fewer vibrations due to the absence of reciprocating movements (Bensinger, 1973). Additionally, its compact and lightweight design allows for greater power density (Froede, 1961), and its ability to reach higher RPMs with ease results in a more consistent power output and an exhilarating driving experience (Yamamoto, 1981). The simplicity of its mechanical design also contributes to lower maintenance requirements (Norbye, 1971), making it particularly suitable for performance-oriented applications (Ludvigsen, 1973). Finally, its continuous combustion cycle enables efficient power delivery, enhancing overall engine responsiveness and smooth acceleration (Kamo, Kakwani, & Hady, 1986).

Academic investigations have focused on enhancing the performance and efficiency of Wankel rotary engines through various innovative approaches. Among these, analytical methods seek theoretical understanding, but the complex kinematics and thermodynamics of the engine have limited the development of general mathematical models. Experimental methods provide valuable operational insights but are costly and restricted to specific geometries. Computational models, meanwhile, enable detailed internal simulations and optimizations, though they are resource-intensive.

For instance, Shen and Costall (2024) employed CFD to analyze combustion in Wankel engines and proposed turbulent jet ignition to mitigate incomplete combustion.

Yang et al. (2022) explored hydrogen-enriched hybrid cycles, noting enhanced performance at high altitudes. Kim, Park, Park, and Lee (2024) applied 1D CFD in GT-POWER to optimize port design and fuel efficiency in a novel rotary engine derived from a gerotor pump.

Other studies include Mukherjee and Seshadri (2022), who quantified intake losses in steam Wankel expanders and their impact on power output, and Chen et al. (2020), who built control-oriented models for the 225CS engine using hybrid analytical-neural approaches. Bailey, Turner, and Brace (2018) analyzed direct injection limitations and fuel economy improvements in Wankel engines using one-dimensional modeling.

Technological advances continue with Mazda reintroducing rotary engines in 2023, LiquidPiston's XTS-210 model targeting military applications, and AIE and Freedom Motors refining air-cooled designs and low-emission engines. These efforts demonstrate renewed interest in rotary technology, emphasizing compactness, efficiency, and cleaner fuel options.

Despite these contributions, most research remains fragmented, focusing on isolated subsystems. There remains a need for integrative studies addressing geometry, combustion chemistry, and thermodynamic cycles within a unified modeling framework.

2. Methodology

This section details the mathematical and physical methods used to describe, simulate, and analyze the behavior of a four-stroke Wankel rotary engine. The methodology is structured into four main components: (1) geometric formulation, (2) thermodynamic cycle modeling, (3) combustion process modeling using the SIR framework, and (4) torque and power analysis using vectorial mechanics. Together, these components provide a comprehensive framework to understand the performance, efficiency, and dynamic response of the engine under varying operating conditions.

2.1 Geometric Formulation

2.1.1 Epitrochoidal Curve of the Stator. The internal shape of the stator is defined by an epitrochoidal curve, governed by the following parametric equations Yamamoto (1981):

$$x = e \cdot \cos(n\theta) + R \cdot \cos(\theta) \quad (1)$$

$$y = e \cdot \sin(n\theta) + R \cdot \sin(\theta) \quad (2)$$

where e is the eccentricity, R is the radius of the base circle, and θ is the rotation angle. This curve defines the trajectory of the rotor center and the envelope of the rotor's movement.

2.1.2 Family of Curves Composing the Rotor. The rotor's profile ensures continuous contact with the epitrochoidal stator. Its geometry is described by the following equations Yamamoto (1981):

$$X = R \cos(2v) + \frac{3}{2} \frac{e^2}{R} (\cos(8v) - \cos(4v)) + e \sqrt{1 - \frac{9e^2}{R^2} \sin^2(3v)} \cdot (\cos(5v) + \cos(v)) \quad (3)$$

$$Y = R \sin(2v) + \frac{3}{2} \frac{e^2}{R} (\cos(8v) + \sin(4v)) + e \sqrt{1 - \frac{9e^2}{R^2} \sin^2(3v)} \cdot (\sin(5v) - \sin(v)) \quad (4)$$

These equations are key to ensuring geometric compatibility and proper sealing during rotation.

2.1.3 Volume Functions of the Chambers. The instantaneous volume of each chamber varies with the rotor angle. The three chamber volumes are given by:

$$V_1 = \left[\left(e^2 + \frac{R^2}{3} \right) \pi - \frac{\sqrt{3}}{4} R^2 \right] W - \frac{3\sqrt{3}}{2} W e R \sin \left(2 \left(\theta - \frac{2\pi}{3} \right) + \frac{\pi}{6} \right) \quad (5)$$

$$V_2 = \left[\left(e^2 + \frac{R^2}{3} \right) \pi - \frac{\sqrt{3}}{4} R^2 \right] W - \frac{3\sqrt{3}}{2} W e R \sin \left(2\theta + \frac{\pi}{6} \right) \quad (6)$$

$$V_3 = \left[\left(e^2 + \frac{R^2}{3} \right) \pi - \frac{\sqrt{3}}{4} R^2 \right] W - \frac{3\sqrt{3}}{2} W e R \sin \left(2 \left(\theta + \frac{2\pi}{3} \right) + \frac{\pi}{6} \right) \quad (7)$$

These functions form the basis for the thermodynamic analysis of the Otto cycle.

2.2 Thermodynamic Modeling of the Otto Cycle

2.2.1 Polytropic Compression and Expansion. Using a polytropic relation for adiabatic processes:

$$P_2 = P_1 \left(\frac{V_1}{V_2} \right)^\gamma \quad (8)$$

this equation is applied during the compression and expansion phases to estimate pressure evolution based on volume changes defined in the previous subsection.

2.2.2 Heat Addition (Isochoric Process). Assuming constant volume during combustion, the final temperature is calculated using:

$$T_3 = \frac{Q}{m_{\text{total}} \cdot C_v} + T_2 \quad (9)$$

and the corresponding pressure is derived from:

$$P_3 = P_2 \cdot \frac{T_3}{T_2} \quad (10)$$

The composition of the mixture, based on the air-fuel ratio (AFR), determines the molar mass and heat capacity of the reacting gases.

2.3 Torque and Power Analysis via Vectorial Mechanics

2.3.1 Vectorial Position Analysis of the Rotor. To complement the thermodynamic and geometric modeling, a vector-based mechanical analysis approach was implemented to analyze the position of key points on the rotor throughout the engine cycle. This method provides a dynamic view of the forces and torques generated by combustion and how they act on the rotor.

The position of the rotor's centroid $\vec{C}_r$ at each angular step θ is given by:

$$\vec{C}_r(\theta) = [e \cdot \cos(\theta), e \cdot \sin(\theta)] \quad (11)$$

where e is the eccentricity of the rotor's path.

The angular position of the rotor itself is governed by the relation $\alpha(\theta) = \frac{1}{3}\theta$, which ensures proper phasing relative to the housing epitrochoid. At each step, the rotor triangle is rotated and translated, and the center of one of its sides (e.g., side AB) is tracked.

Two vectors are calculated for each frame:

- $\vec{r}_1$: from the rotor centroid to the center of side AB.
- $\vec{r}_2$: from the engine axis (origin) to the rotor centroid.

The total position vector of the point of application of force is:

$$\vec{r}_{\text{total}} = \vec{r}_1 + \vec{r}_2, \quad (12)$$

providing the arm for torque calculation. As shown in the figure

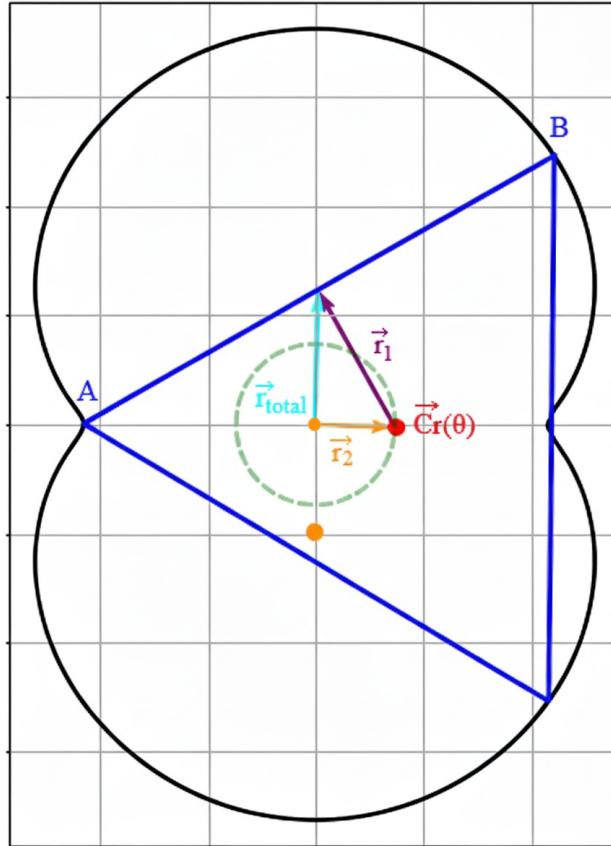


Figure 1. Representation of the previously mentioned vectors in the wankel engine.

The direction of the applied force is considered opposite to $\vec{r}_1$, representing the pressure exerted on the rotor face during expansion. This direction is normalized to obtain a unit vector $\hat{f}$, and the instantaneous torque is then calculated as a cross product:

$$\tau(\theta) = \vec{r}_{\text{total}}(\theta) \times \left(F(\theta) \cdot \hat{f}(\theta) \right), \quad (13)$$

where $F(\theta)$ is the force derived from pressure multiplied by the chamber area at that moment.

This approach enables a dynamic, time-resolved calculation of torque, revealing asymmetries and time dependencies within the engine cycle that are not visible from thermodynamic equations alone. The methodology bridges geometric modeling and dynamic vector analysis, improving the accuracy of performance simulations and enabling optimization based on vectorial torque profiles.

To analyze the mechanical response, the torque τ is computed using the vector cross product between the arm of force $\vec{r}$ (from the origin to the point of force application) and the force vector $\vec{F}$:

$$\tau = \vec{r} \times \vec{F} = r_x F_y - r_y F_x \quad (14)$$

The direction of the force is assumed opposite to the vector connecting the rotor center to the point of application. The torque is computed at each time step using the pressure-based force and the position vector of the rotor.

Power output is computed as:

$$P = \tau \cdot \omega \quad (15)$$

and the net mechanical work is estimated from:

$$W_{\text{net}} = \oint P dV \quad (16)$$

This analysis helps characterize both instantaneous and average performance of the engine.

3. Results

3.1 Geometrical Validation of the Rotor and Chamber

The geometry of the rotor and housing was first implemented using parametric equations derived from the epitrochoidal path and the kinematic behavior of the Wankel mechanism. The resulting rotor shape and apex trajectory were validated graphically, confirming their periodic and symmetric nature. This validation was further supported by simulating the rotor's motion and comparing the results with the analytical representation obtained from Equations (1) and (2). This step provided a reliable geometric foundation for the subsequent volume and thermodynamic analyses.

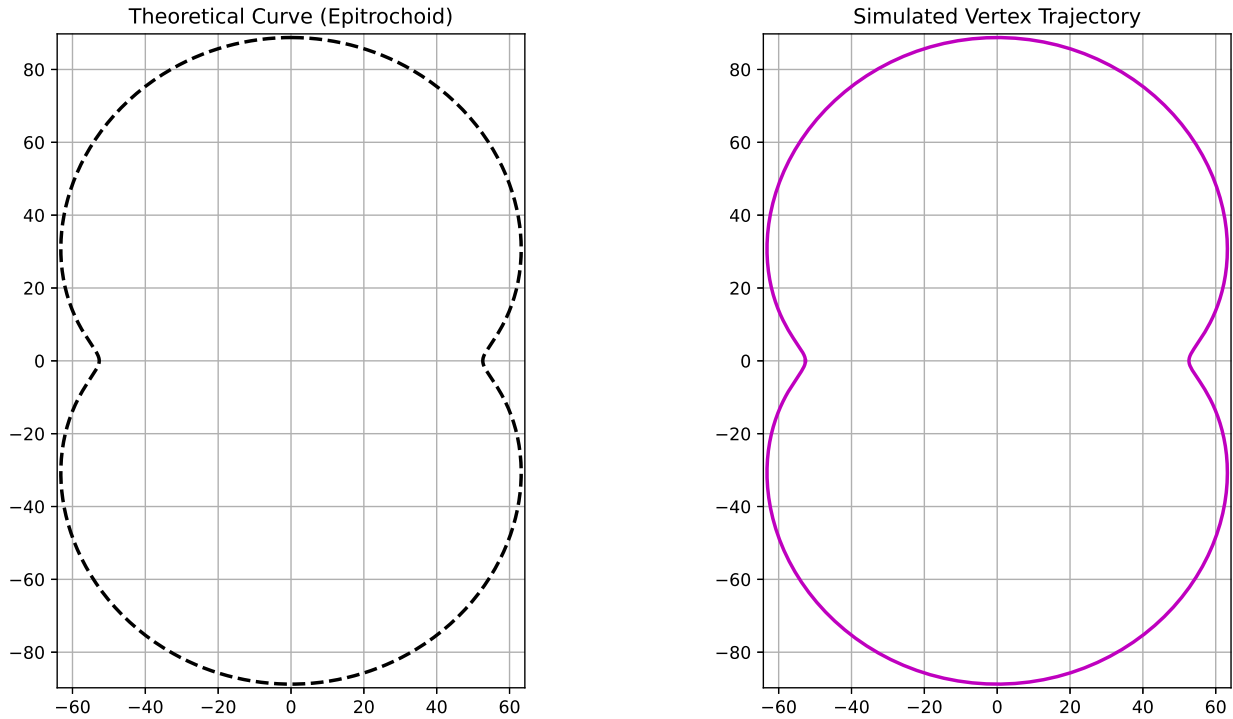


Figure 2. Comparison between the theoretical epitrochoid (left), obtained from the analytical formulation given by Equations (1) and (2), using parameters $R = 70.71$ and $e = 10.06$ with the side of the triangle for the simulation $L = R\sqrt{3}$ and the simulated vertex trajectory (right). The close match between both curves validates the geometric implementation of the rotor's motion and confirms the accuracy of the kinematic model.

3.2 Geometrical Simulation

By varying key geometric parameters of the rotary Wankel engine, such as the rotor radius R and the eccentricity e , it is possible to obtain different rotor–housing configurations that directly influence the shape of the combustion chambers. These geometric changes modify the maximum and minimum volumes reached during the engine cycle, thereby altering the compression ratio. Increasing e , for example, typically reduces the minimum chamber volume, leading to a higher compression ratio, while changing R can affect both the displacement and the overall chamber geometry. As a result, adjusting these parameters allows for the design of engines with tailored performance characteristics, enabling optimization for power output, efficiency, or specific operating conditions.

In this study, three different geometries will be compared, starting from Geometry 1 as the baseline. The variations will be introduced by modifying the parameters R and e , allowing the influence of these geometric changes on the engine’s performance to be analyzed

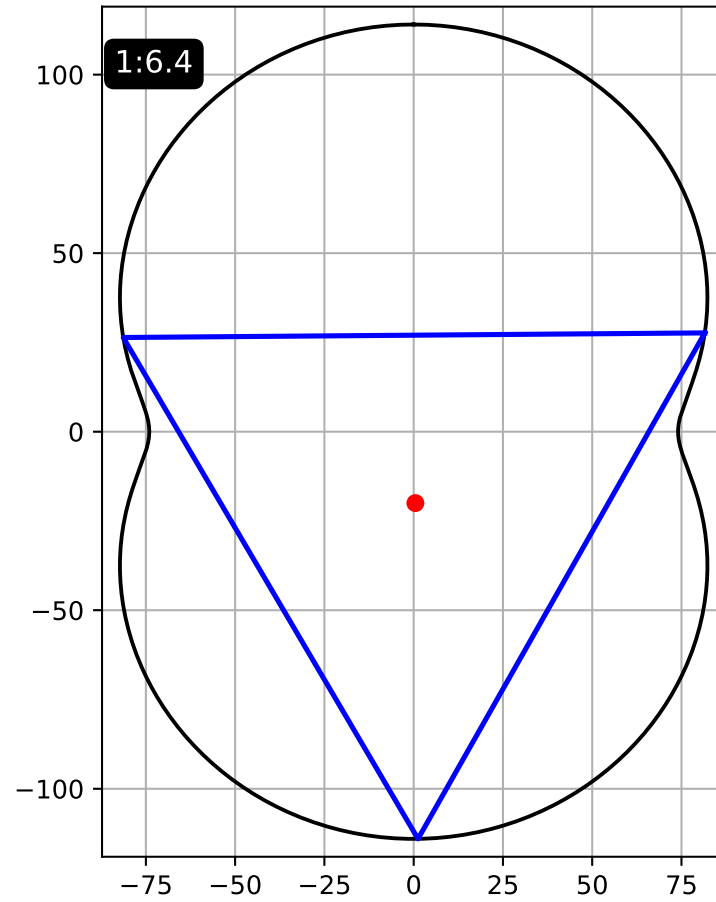


Figure 3. Geometry 1 with a compression ratio of 1:6.4, with parameters $R = 94.04$ mm, $e = 20$ mm, and $W = 30$ mm.

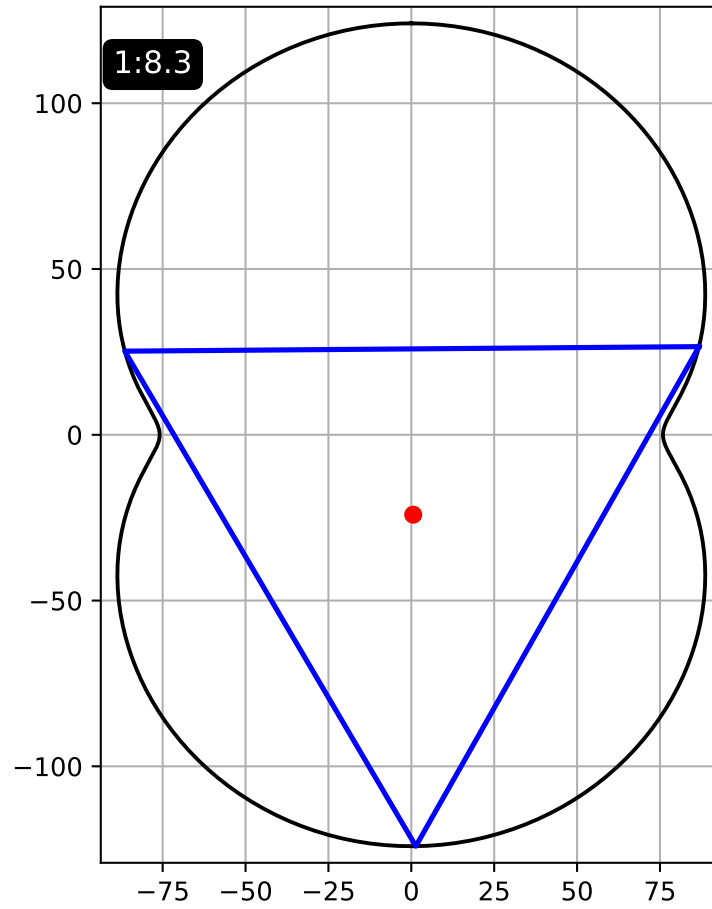


Figure 4. Geometry 2 with a compression ratio of 1:8.3, with parameters $R = 100$ mm, $e = 24.1$ mm, and $W = 30$ mm.

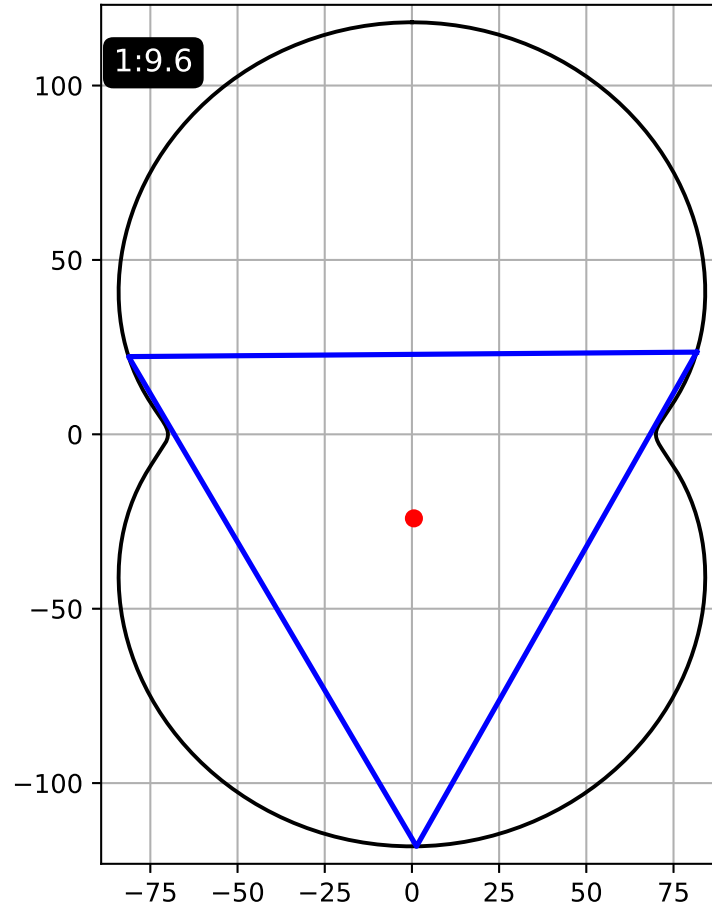


Figure 5. Geometry 3 with a compression ratio of 1:9.6, with parameters $R = 94.04$ mm, $e = 24.1$ mm, and $W = 30$ mm.

The following results present the comparison of three different geometries of the rotary chamber. Pressure and temperature profiles are shown as functions of the crank angle for each configuration, allowing us to observe how the geometric parameters influence the thermodynamic behavior during compression and expansion. In addition, a P-V diagram is provided for the three geometries using gasoline as the fuel, which highlights the differences in the enclosed cycle area and therefore the work output of each case.

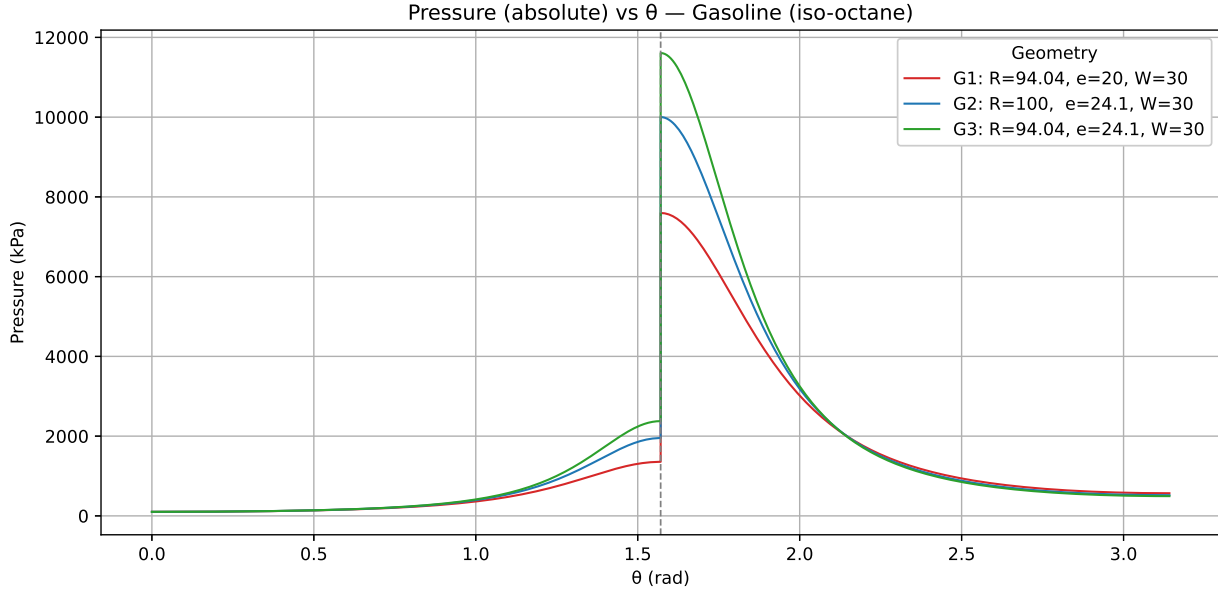


Figure 6. Pressure plot for the three geometries.

The pressure curves for the three geometries highlight the strong influence of the eccentricity e and the radius R on the peak values reached during the cycle. Geometry G1 ($R = 94.04$ mm, $e = 20$ mm, $W = 30$ mm) shows the lowest maximum pressure, indicating a lower compression effect compared to the other two cases. Geometry G2 ($R = 100$ mm, $e = 24.1$ mm, $W = 30$ mm) achieves a higher peak pressure due to the larger chamber radius combined with a greater eccentricity, which increases the compression ratio. Geometry G3 ($R = 94.04$ mm, $e = 24.1$ mm, $W = 30$ mm) reaches the highest peak pressure of all, showing that eccentricity is a critical parameter in enhancing the compression and the resulting pressure rise after combustion.

Despite these differences in maximum pressure, all geometries show a similar qualitative trend: a gradual pressure increase during compression, a sudden rise due to isochoric heat addition, and a subsequent decrease during expansion. The magnitude of the peak pressure directly affects the indicated work per cycle, suggesting that G3 would deliver the highest power output, followed by G2, while G1 would provide the lowest.

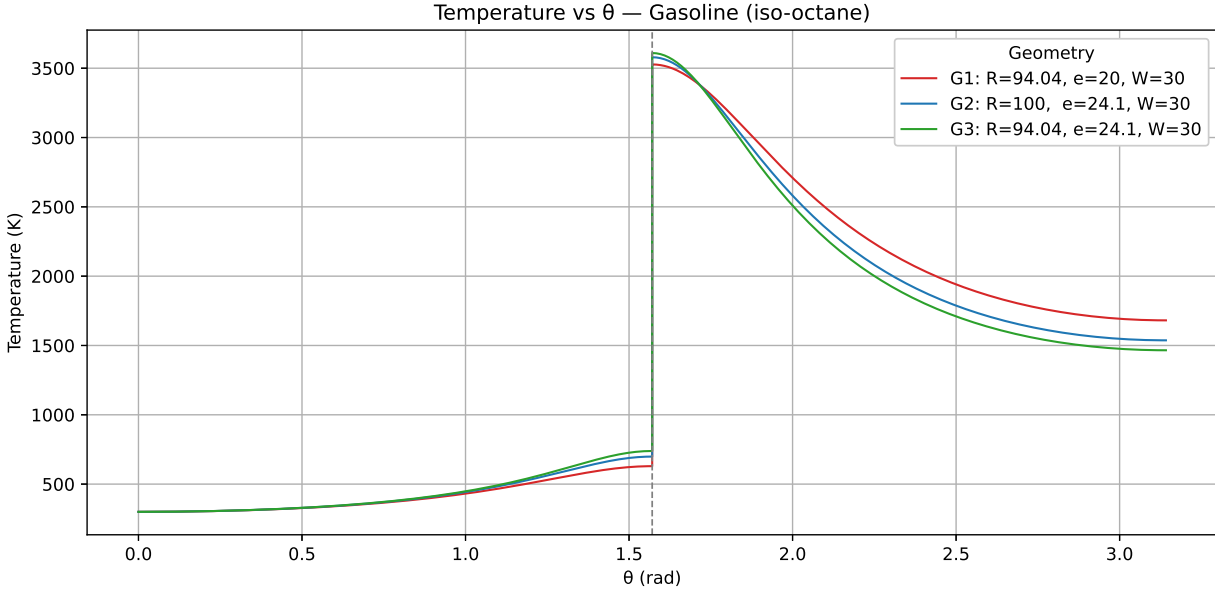


Figure 7. Temperature plot for the three geometries.

The temperature profiles for the three geometries show the typical evolution of the cycle: a moderate rise during compression, a sharp jump after heat release, and a smooth decrease during expansion. The differences among the geometries are mainly observed in the magnitude of the peak temperature.

Geometry G3 ($R = 94.04$ mm, $e = 24.1$ mm, $W = 30$ mm) produces the highest maximum temperature. This is because the lower eccentricity reduces the effective chamber volume at compression, increasing the temperature rise when heat is released.

Geometry G2 ($R = 100$ mm, $e = 24.1$ mm, $W = 30$ mm) shows slightly lower peak temperatures than G1. The larger rotor radius combined with higher eccentricity increases the chamber volume, which spreads the added heat over more mass, lowering the temperature peak.

Geometry G1 ($R = 94.04$ mm, $e = 20$ mm, $W = 30$ mm) reaches the lowest maximum temperature of the three. The combination of smaller radius with greater eccentricity further enlarges the compressed volume, reducing the intensity of the temperature rise.

Overall, while the three geometries exhibit the same qualitative behavior, peak

temperature decreases as eccentricity and chamber volume increase, highlighting how geometric design strongly influences the thermal conditions of the working fluid.

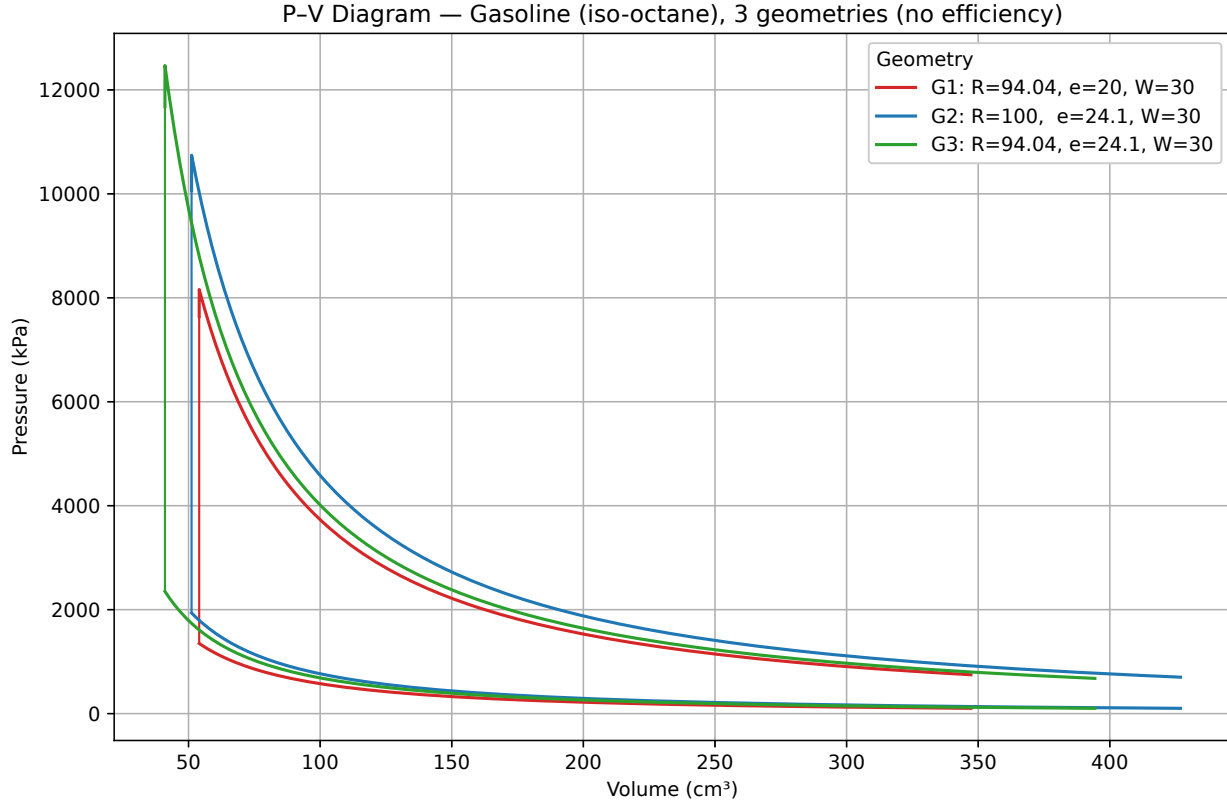


Figure 8. P-V plot for the three geometries.

The P–V diagram clearly shows how geometry affects the thermodynamic cycle of the rotary chamber. While all three configurations follow the same qualitative behavior—compression, isochoric heat addition, and expansion—the enclosed cycle areas and peak pressures differ significantly.

For **G1** ($R = 94.04$ mm, $e = 20$ mm, $W = 30$ mm), the cycle exhibits the lowest work, $W_{\text{net}} = 0.737$ kJ, together with the highest specific fuel consumption, $\text{SFC} = 0.038$ g/kJ. The corresponding peak values are $P_{\text{max}} = 8158.78$ kPa and $T_{\text{max}} = 3521.67$ K. In **G2** ($R = 100$ mm, $e = 24.1$ mm, $W = 30$ mm), the net work rises to 1.024 kJ (+38.9% relative to G1), while SFC decreases to 0.033 g/kJ (−13.2%). Peak pressure reaches 10742.99 kPa (+31.7%) with $T_{\text{max}} = 3570.52$ K (+1.39%). Finally, **G3**

($R = 94.04$ mm, $e = 24.1$ mm, $W = 30$ mm) sustains high work, 1.005 kJ (+36.4% vs G1), and achieves the lowest SFC, 0.031 g/kJ (−18.4%), while exhibiting the highest peak pressure, $P_{\max} = 12467.94$ kPa (+52.8% vs G1), and $T_{\max} = 3598.84$ K (+2.19%).

Comparatively, increasing eccentricity from $e = 20$ to $e = 24.1$ at constant R (G1 $\rightarrow$ G3) is associated with simultaneous gains in W_{net} and reductions in SFC, at the expense of a pronounced rise in $P_{\max}$, while $T_{\max}$ changes little (within $\leq 2.2\%$). Increasing radius from $R = 94.04$ mm to $R = 100$ mm at constant e (G3 $\rightarrow$ G2) moderates $P_{\max}$ by about 14% relative to G3, maintains a similarly high W_{net} , and incurs only a slight SFC penalty.

Overall, the comparison demonstrates that increasing eccentricity (e) raises the compression ratio and cycle efficiency, while increasing radius (R) enlarges volume but slightly reduces the maximum pressure. Thus, G3 offers the most favorable balance for maximizing indicated work output, followed by G2, with G1 being the least effective.

3.3 Simulation of different fuels

The numerical model developed in this study provides a detailed representation of the thermodynamic behavior of the rotary Wankel engine. The results include the evolution of pressure during compression and expansion, as well as the corresponding temperature changes. These outputs were computed for a defined engine configuration, considering ambient conditions and the geometric parameters $R = 94.04$ mm, $e = 24.1$ mm, $W = 30$ mm, and a compression ratio of 1:9.6. Furthermore, three different fuels were used to compare the thermodynamic cycle: gasoline, methane, and bioethanol. To develop this thermodynamic study, the lower heating values (LHV) of the fuels were considered: gasoline 41.1 MJ kg^{−1} Davis, Diegel, and Boundy (2019), methane 50.0 MJ kg^{−1} Turns (2013), and bioethanol 26.8 MJ kg^{−1} Demirbas (2008). These values were used to calculate the energy released per gram of fuel in the combustion chamber of the Wankel engine.

From the chamber volume functions presented in Equations (5), (6), and (7), and by applying the thermodynamic Equation (8) for the compression process, Equations (9) and (10) for the heat addition phase, together with Equation (11) and the SIR combustion

model described in Equation (12), it is possible to calculate the variations of pressure and temperature within the combustion chamber of the Wankel engine during both the compression and expansion phases. This integrated approach links the geometric evolution of the chamber with the thermodynamic processes, providing a complete description of the working engine cycle.

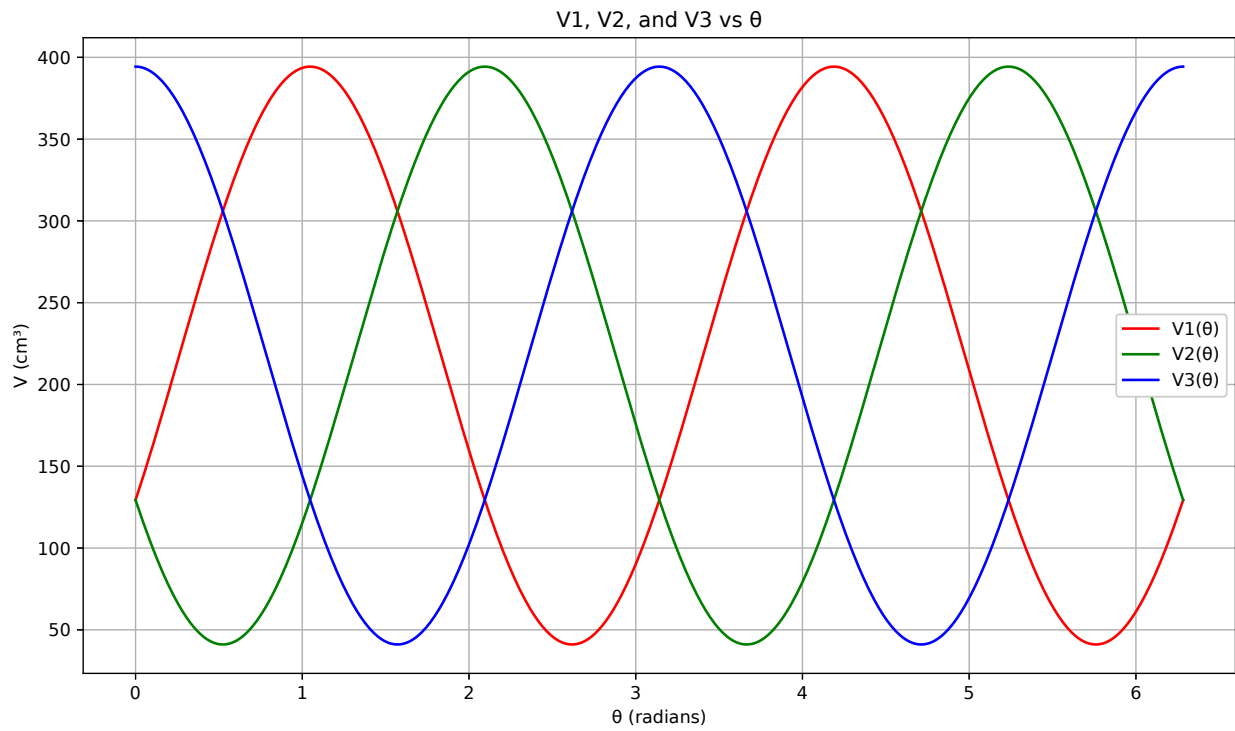


Figure 9. Volume of the engine chambers with respect to the rotor angle.

The following figures illustrate the behavior of pressure, temperature, and the P–V diagram for three different fuels – gasoline (iso-octane), methane (CH₄), and bioethanol (CH₃CHOH) – considering the previously defined rotor and housing geometry.

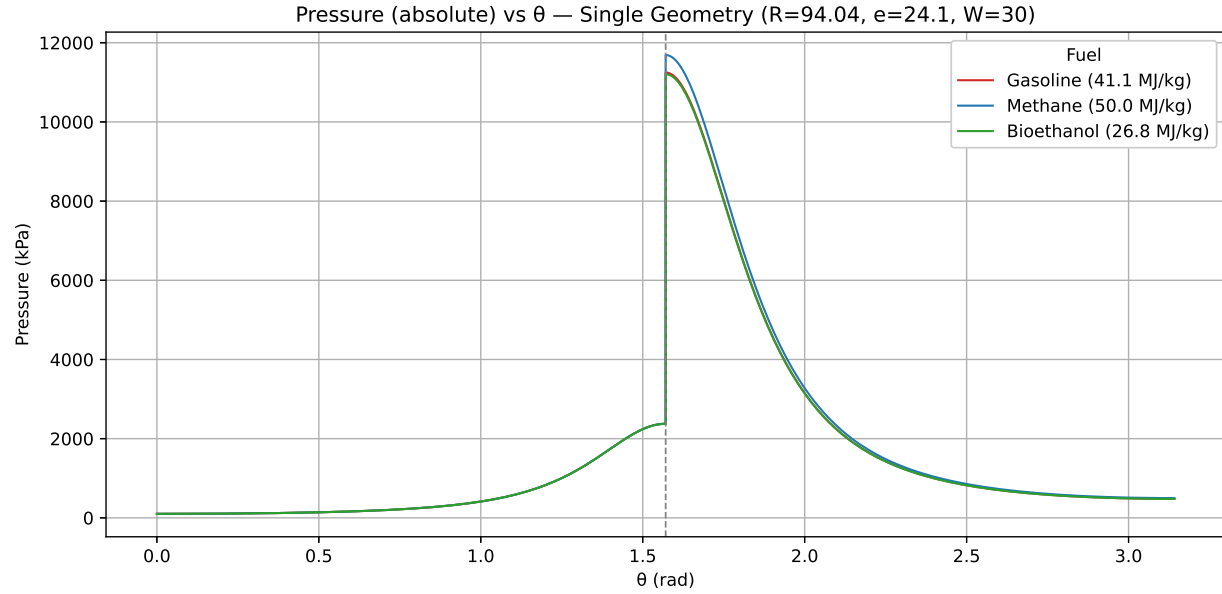


Figure 10. Pressure during the compression and expansion phases of the engine for 3 different fuels.

The three pressure traces follow the same pattern: a gradual rise during compression, a sharp jump at the heat-release point, and a smooth decay through expansion. In this comparison, methane shows the highest peak pressure, gasoline is slightly lower, and bioethanol is the lowest. Methane’s peak is highest because, even though its stoichiometric mixture uses less fuel per unit air, the mixture’s effective heat capacity is lower and the heat release raises temperature—and thus pressure—more sharply. Gasoline lands close behind due to strong heating but a slightly “heavier” mixture that tempers the temperature rise. Bioethanol trails because its lower heating value and richer mixture (plus higher specific heat of the fuel) spread the added energy over a more heat-absorbent charge, yielding a smaller pressure spike. The overall curve shapes are similar across fuels; what changes most is the magnitude of the peak.

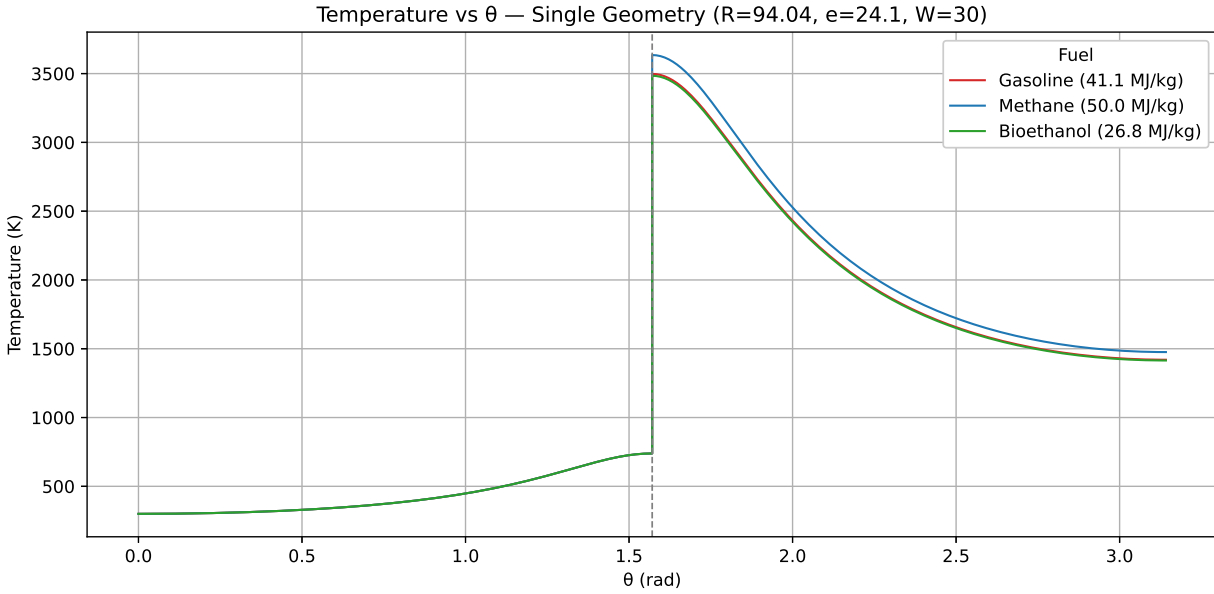


Figure 11. Temperature during the compression and expansion phases of the engine for 3 different fuels.

The temperature traces for gasoline, methane, and bioethanol share the same shape: a steady rise during compression, a sharp step at the isochoric heat-release point (near $\pi/2$), and a monotonic drop through expansion. The peak ordering is Gasoline $>$ Methane $>$ Bioethanol. Gasoline attains the highest maximum because, for the same trapped air mass, its combination of heating value and mixture properties yields the largest effective sensible heating at constant volume. Methane peaks slightly lower—its leaner stoichiometric mixture means less fuel mass per unit air, moderating the temperature jump. Bioethanol shows the lowest peak, consistent with its lower LHV and higher specific heat, which increase the mixture’s heat-absorbing capacity and limit the rise.

After the peak, all three curves cool at comparable rates as the volume increases and the gases expand adiabatically. The differences among fuels are therefore most visible at and immediately after the heat-addition event; away from that region, the curves converge, underscoring that fuel choice primarily shifts the magnitude of peak temperature rather than the overall trend.

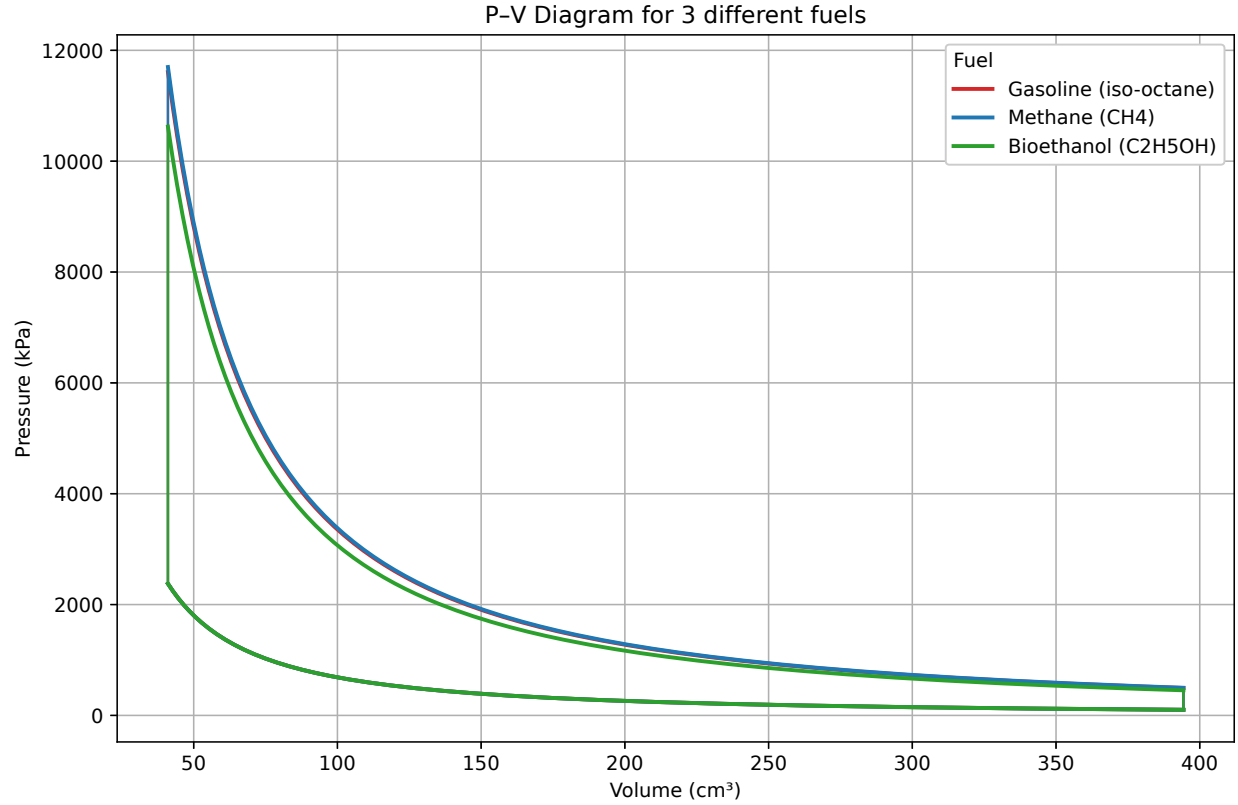


Figure 12. P-V diagram of the cycle with 3 different fuels.

The P–V diagram compares the thermodynamic performance of gasoline (iso-octane), methane (CH_4), and bioethanol ($\text{C}_2\text{H}_5\text{OH}$) under identical geometric conditions. While the overall cycle shapes remain similar, the peak pressures, maximum temperatures, and specific fuel consumptions highlight the distinct characteristics of each fuel.

Methane achieves the highest final pressure after the isochoric process (**11,699.19 kPa**) and a peak temperature of about **3637.88 K**. This leads to the largest work output per cycle (**0.865 kJ**) and the lowest specific fuel consumption (**0.031 g/kJ**). These results confirm methane’s superior thermodynamic efficiency in this configuration.

Gasoline (iso-octane) exhibits a final pressure of **11,607.89 kPa** and a maximum temperature of **3609.49 K**. Its net work per cycle is slightly lower than methane at **0.859 kJ**, with a higher specific fuel consumption (**0.037 g/kJ**). This positions gasoline as the second-best performer, close to methane in terms of work but less favorable in fuel

efficiency.

Bioethanol displays the lowest final pressure (**10,623.15 kPa**) and temperature (**3383.28 K**) among the three fuels. Consequently, its net work per cycle (**0.799 kJ**) is the smallest, and its specific fuel consumption (**0.065 g/kJ**) is significantly higher. These results underscore the reduced energy density of bioethanol and its comparative disadvantage in efficiency under the same operating conditions.

Because this study relies on a cycle-level thermodynamic simulation without a coupled load or dynamometer model, we cannot report SBHP (brake specific horsepower). SBHP is derived from brake power, which requires shaft torque under an externally applied load and must account for mechanical, friction, and pumping losses. Our model does not simulate the load path or include a validated loss map; therefore brake torque—and thus brake power and SBHP—cannot be resolved with sufficient fidelity. Any SBHP value would be speculative. Instead, we report indicated quantities (e.g., IMEP, indicated work/power) and theoretical torque; obtaining SBHP would require experimental dynamometer data or an extended model that explicitly includes a load and calibrated mechanical losses.

Overall, the diagram and data confirm that methane provides the best performance in terms of pressure, work output, and fuel efficiency, followed closely by gasoline, while bioethanol, although cleaner and renewable, delivers less energy per cycle and requires more fuel mass for the same output.

3.4 Vectorial Torque Analysis

Torque was computed from the normal force on the active rotor side and the lever arm with respect to the origin,

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}, \quad \mathbf{F} = (P(\theta) - P_{\text{atm}}) \mathbf{A} \quad A = L W, \quad (17)$$

where A is the effective thrust area in m^2 , $L = \sqrt{3} R$ is the rotor side length, W is the axial width. The pressure trace $P(\theta)$ comes from the thermodynamic cycle model (adiabatic compression, isochoric heat addition, adiabatic expansion).

Torque vs. angle for three geometries

Figure 16 compares three geometries under the same fuel:

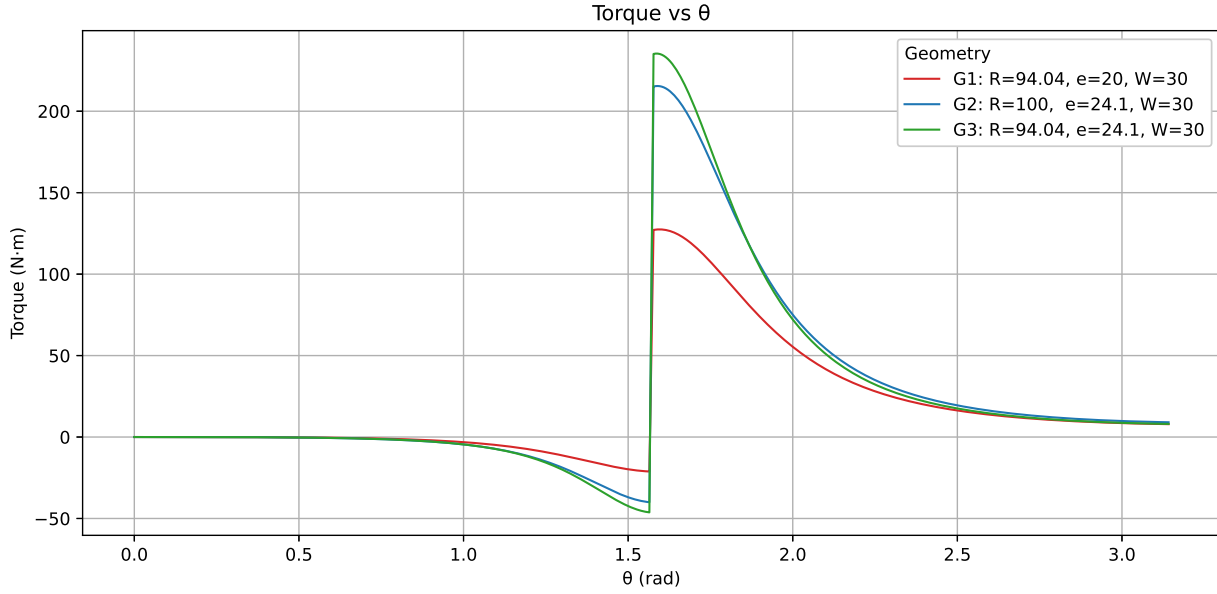


Figure 13. Torque vs. angle for the three geometries.

We compare three geometries under identical operating and fuel assumptions: G1 ($R=94.04$, $e=20$, $W=30$), G2 ($R=100$, $e=24.1$, $W=30$), and G3 ($R=94.04$, $e=24.1$, $W=30$).

Peaks occur just after the isochoric step, near $\theta \approx \pi/2$:

Geometry	Peak torque $\tau_{\max}$ [N·m]	Angle [rad]	vs. G1
G1	127.466	1.59	—
G2	215.482	1.59	+69.1%
G3	235.381	1.59	+84.7%

Relative peak comparisons: G1→G2 +69.1%, G1→G3 +84.7%, G2→G3 +9.2%.

Torque vs. angle for three fuels (fixed geometry)

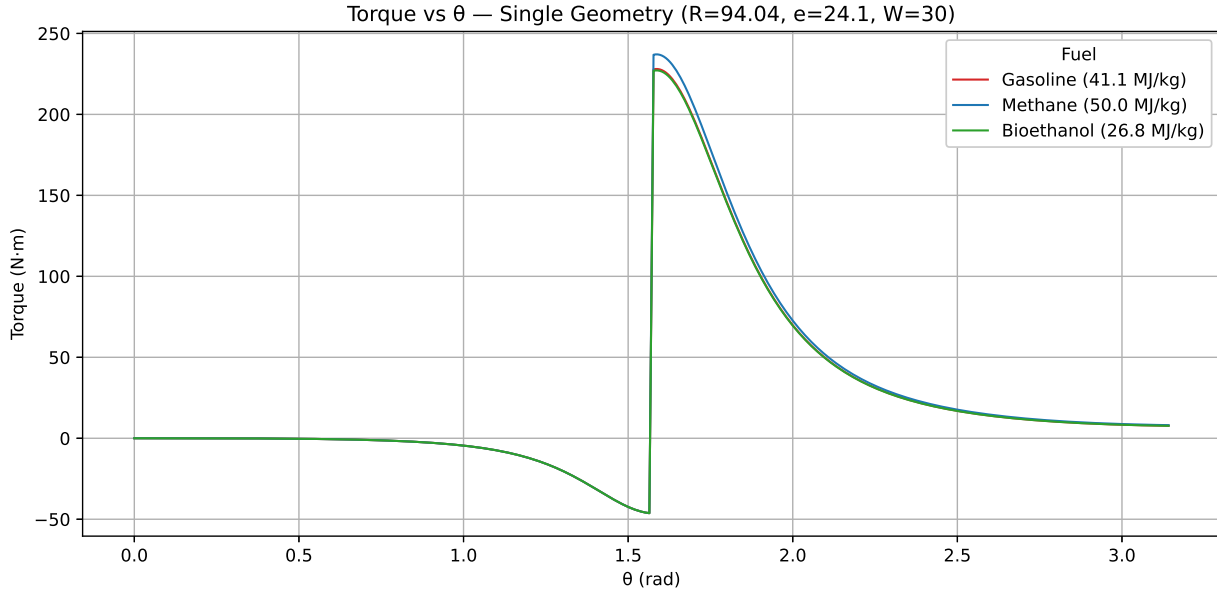


Figure 14. Torque vs. angle for the three different fuels with the same geometry.

Geometry held constant at $R = 94.04$ mm, $e = 24.1$ mm, $W = 30$ mm, and a compression ratio of 1:9.6. We compare *Gasoline* (*iso-octane*), *Methane* (CH_4), and *Bioethanol* (C_2H_5OH) under identical cycle assumptions (adiabatic compression, isochoric heat addition, adiabatic expansion).

Peak torque

Maximum torque from the torque–angle curves:

All fuels reach their maximum torque at essentially the same crank angle ($\theta \approx 90.72^\circ$), just after the isochoric heat addition. The minimum torque occurs slightly before that ($\theta \approx 89.62^\circ$) and is similar for all fuels ($\tau_{\min} \approx -46.27$ N·m). In terms of amplitude, methane shows the highest peak (237.06 N·m), about +4.0% above gasoline (228.03 N·m) and +4.2% above bioethanol (227.14 N·m). The negative torque magnitude during compression is around 20% of each fuel's peak, indicating the same basic pulse shape across fuels; the fuel choice mainly scales the torque level rather than shifting its phase.

Fuel	$\tau_{\max}$ [N·m]	Angle [°]	vs. Gasoline	vs. Methane
Gasoline (iso-octane)	228.032	90.70	—	−3.8%
Methane (CH ₄)	237.056	90.70	+4.0%	—
Bioethanol (C ₂ H ₅ OH)	227.136	90.70	−0.4%	−4.2%

Table 1

Comparison of peak torque ($\tau_{\max}$) and peak angle for each fuel. Relative columns report percentage difference in $\tau_{\max}$.

3.5 Comparison with a real engine

In order to validate the numerical model, a comparison is planned between the simulated engine and a commercially available Wankel rotary engine, the AIE 650S 120 BHP. According to the manufacturer’s datasheet, this single-rotor engine has a displacement of **650 cm³**, delivers up to **120 bhp (90 kW)** of power at **8000 rpm**, and develops a torque of approximately **80 lb·ft**. The engine operates with a compression ratio of **9.6:1**, features a liquid-cooled system (SPARCS*), and supports multiple fuel types including AVGAS, gasoline, and heavy fuels. Its specific fuel consumption ranges between **310 and 350 g/kWh**, and the core engine weight is **28 kg** *650S 120 BHP Wankel Rotary Engine: Specification Datasheet* (2016) .

For the purposes of the simulation, the compression ratio and displacement were matched to the values of the real engine to ensure consistency in the thermodynamic conditions and chamber geometry. By setting these parameters as boundary conditions, the simulated model replicates the fundamental operating characteristics of the 650S. This alignment allows the subsequent comparative analysis to focus on performance indicators such as pressure evolution, net work per cycle, and specific fuel consumption, while preserving the equivalence in compression and displacement between the virtual and real engines.

To enable a like-for-like comparison with the reference single-rotor engine, the simulated stator-rotor geometry was calibrated using $R = 94.04$ mm, $e = 24.1$ mm, and $W = 30.0$ mm. With these parameters, the chamber-volume function $V(\theta; R, e, W)$ yields the maximum and minimum volumes $V_{\max}$ and $V_{\min}$ such that

$$\text{CR} = \frac{V_{\max}}{V_{\min}} = 9.6.$$

Adopting the standard Wankel convention (single rotor, three working chambers), the nominal engine displacement is 650 cc class of the real engine. This geometric matching ensures that subsequent differences in performance arise from thermodynamic assumptions rather than gross geometric discrepancies.

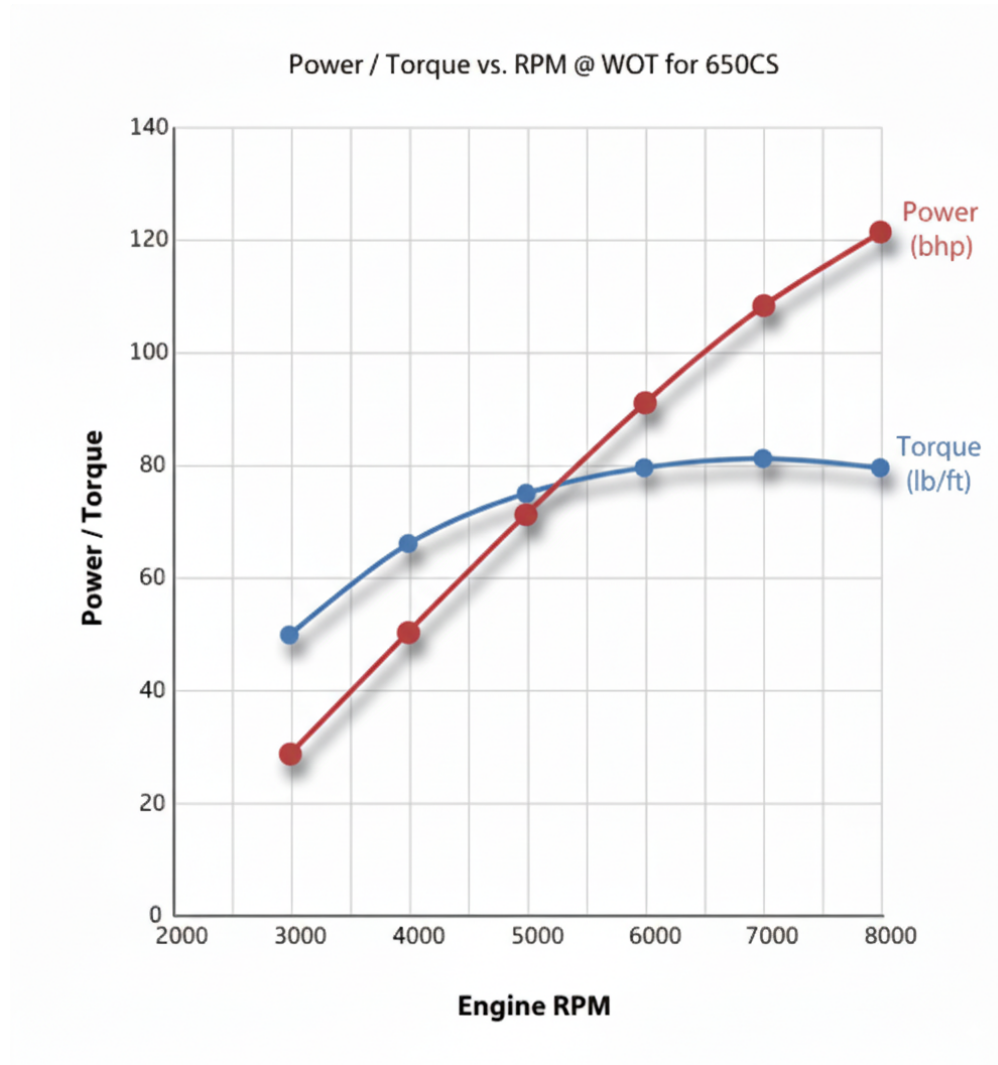


Figure 15. Real Engine Power/Torque vs RPMS.

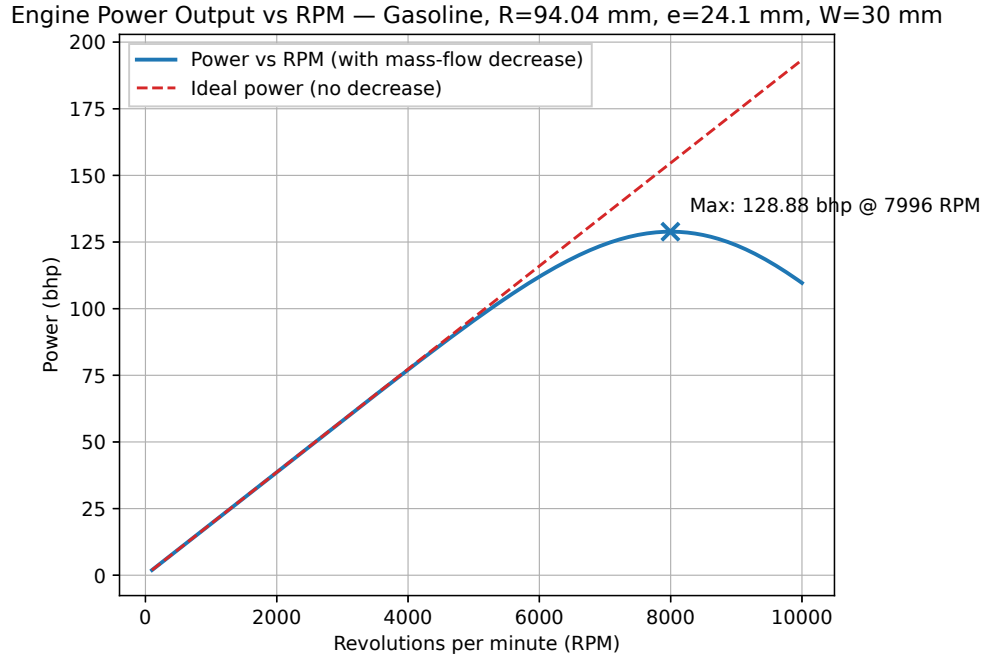


Figure 16. Real Engine Power/Torque vs RPMS.

4. Technical Discussion

The predictive capacity of the model is evaluated, including its ability to replicate key engine behaviors. Advantages of the model over existing approaches are discussed, such as its generality, computational simplicity, or adaptability to new configurations. Limitations are also addressed, particularly regarding idealized assumptions like adiabatic processes or neglect of heat losses. The model's potential application to other engine architectures is briefly explored.

References

- 650s 120 bhp wankel rotary engine: Specification datasheet* (Product datasheet). (2016, January). Lichfield, United Kingdom: Advanced Innovative Engineering (UK) Ltd. Retrieved from <https://www.aieuk.com/> (PDF datasheet, January 2016 revision)
- Bailey, N., Turner, J., & Brace, C. (2018). Comparison of 1-d modelling approaches for wankel engine performance simulation and initial study of the direct injection limitations. *SAE Technical Paper*(2018-01-1452). doi: 10.4271/2018-01-1452
- Bensing, W.-D. (1973). *Rotationskolben-verbrennungsmotoren*. Springer-Verlag.
- Chen, A. S., Vorraro, G., Turner, M., Islam, R., Herrmann, G., Burgess, S., . . . Turner, J. (2020). Control-oriented modelling of a wankel rotary engine: A synthesis approach of state space and neural networks. *SAE Technical Paper*(2020-01-0253). doi: 10.4271/2020-01-0253
- Davis, S. C., Diegel, S. W., & Boundy, R. G. (2019). *Transportation energy data book: Edition 37* (Tech. Rep. No. ORNL/TM-2018/104). Oak Ridge, TN: Oak Ridge National Laboratory, U.S. Department of Energy. Retrieved from <https://tedb.ornl.gov/> (Gasoline lower heating value (LHV) ≈ 41.1 MJ/kg)
- Demirbas, A. (2008). Biofuels sources, biofuel policy, biofuel economy and global biofuel projections. *Energy Conversion and Management*, 49(8), 2106–2116. (Ethanol lower heating value (LHV) ≈ 26.8 MJ/kg) doi: 10.1016/j.enconman.2008.02.020
- Froede, W. G. (1961). The nsu-wankel rotating combustion engine. *SAE Technical Paper*(610017). doi: 10.4271/610017
- Kamo, R., Kakwani, R. M., & Hady, W. (1986). Adiabatic wankel type rotary engine. *SAE Technical Paper*(860616). doi: 10.4271/860616
- Kim, Y.-J., Park, Y.-J., Park, T.-J., & Lee, C.-E. (2024). Development of a simulation model for a new rotary engine to optimize port location and operating conditions using gt-power. *Energies*, 17(18), 4732. Retrieved from <https://www.mdpi.com/1996-1073/17/18/4732> doi: 10.3390/en17184732

- Ludvigsen, K. (1973). *Wankel engines a to z*. SAE International.
- Mukherjee, A., & Seshadri, S. (2022). Numerical investigation of pressure losses and its effect during intake in a steam wankel expander. *arXiv preprint arXiv:2208.13773*. Retrieved from <https://arxiv.org/abs/2208.13773>
- Norbye, J. P. (1971). *The wankel engine: Design, development, applications*. Chilton Book Company.
- Shen, X., & Costall, A. W. (2024). Enhancing the efficiency of rotary thermal propulsion systems. *Energies*, 17(9), 2112. Retrieved from <https://www.mdpi.com/1996-1073/17/9/2112> doi: 10.3390/en17092112
- Turns, S. R. (2013). *An introduction to combustion: Concepts and applications* (3rd ed.). McGraw-Hill. (Methane lower heating value (LHV) ≈ 50.0 MJ/kg)
- Wang, Z., Shuai, S., Li, Z., & Yu, W. (2021). A review of energy loss reduction technologies for internal combustion engines to improve brake thermal efficiency. *Energies*, 14(20), 6656. Retrieved from <https://www.mdpi.com/1996-1073/14/20/6656> doi: 10.3390/en14206656
- Yamamoto, K. (1981). *Rotary engine*. Tokyo: Sankaido.
- Yang, Z., Du, Y., Geng, Q., Gao, X., Er, H., Liu, Y., & He, G. (2022). Performance analysis of a hydrogen-doped high-efficiency hybrid cycle rotary engine in high-altitude environments based on a single-zone model. *Energies*, 15(21), 7948. Retrieved from <https://www.mdpi.com/1996-1073/15/21/7948> doi: 10.3390/en15217948